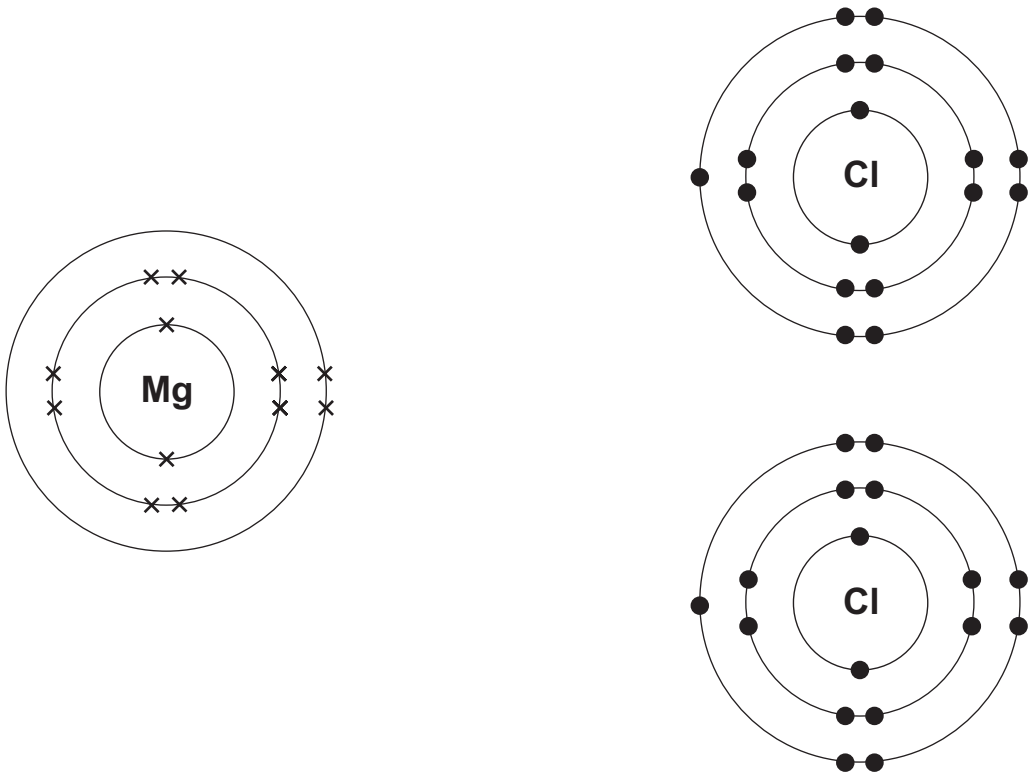


3. (a) Magnesium reacts with chlorine to form magnesium chloride.

The following diagram shows the electronic structures of magnesium and chlorine atoms.



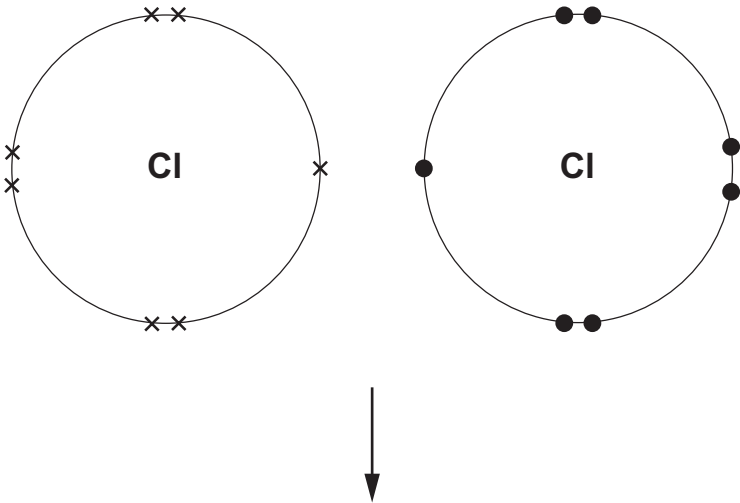
- (i) **Draw arrows on the diagram** to show how electrons are transferred between the magnesium and chlorine atoms during the formation of magnesium chloride. [1]
- (ii) **Complete the table** giving information about the charge and electronic structure of the magnesium and chloride **ions** that are formed. [2]

Ion	Charge	Electronic structure
magnesium	+2	
chloride		2,8,8



(b) Chlorine gas, Cl_2 , consists of two chlorine atoms bonded together.

(i) Draw a diagram in the box to show how the atoms bond to form a chlorine molecule. [1]



(ii) Give the name of this type of bonding.

.....

[1]

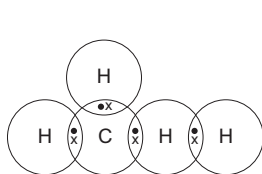
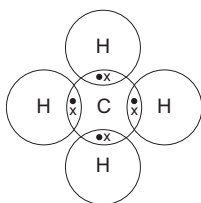
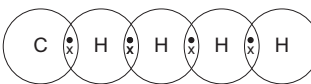
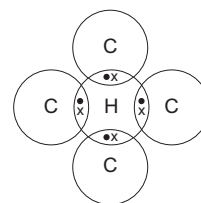
5



3. (a) Methane gas, CH_4 , contains the elements carbon and hydrogen.

Element	Electronic structure
carbon	2,4
hydrogen	1

- (i) I. Put a **tick** (✓) in the box below the diagram which shows how the atoms are bonded in one molecule of methane. [1]


☐

☐

☐

☐

- II. **Circle** the name given to the type of bonding found in methane. [1]

covalent

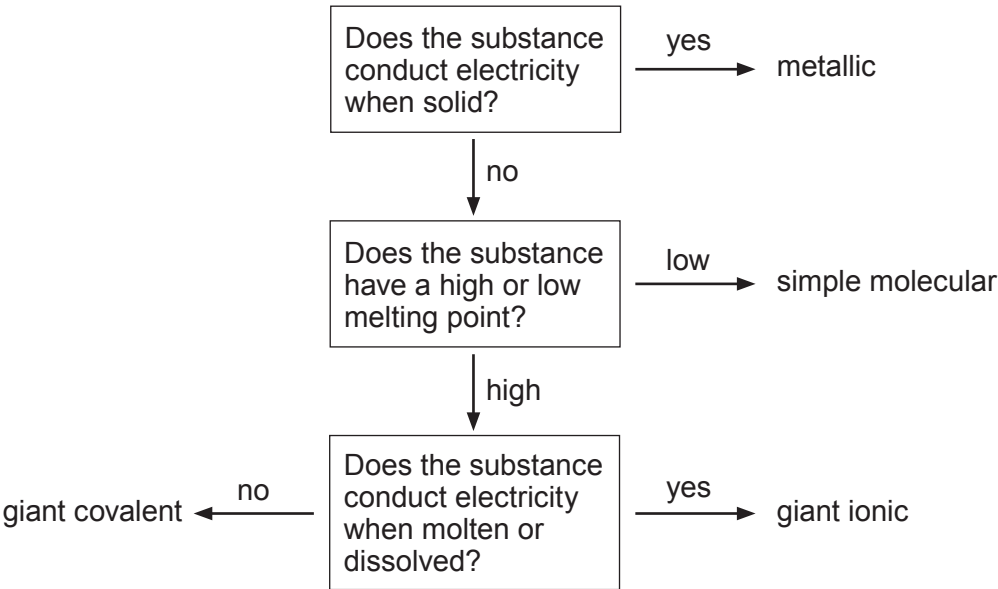
ionic

giant covalent

metallic



(ii) The flow chart below can be used to identify the type of structure found in different substances.



I. Use the flow chart to complete the table. [1]

Substance	Melting point (°C)	Electrical conductivity	Type of structure
A	2072	conducts only when molten	
B	-182	does not conduct electricity	
C	1610	does not conduct electricity	

II. Give the **letter** of the substance, **A**, **B** or **C**, that is most likely to be methane. [1]

Substance

3430U501
09



- (b) Butene, C_4H_8 , also contains the elements carbon and hydrogen. Calculate the percentage by mass of hydrogen in butene. [2]

$$A_r(H) = 1 \quad A_r(C) = 12$$

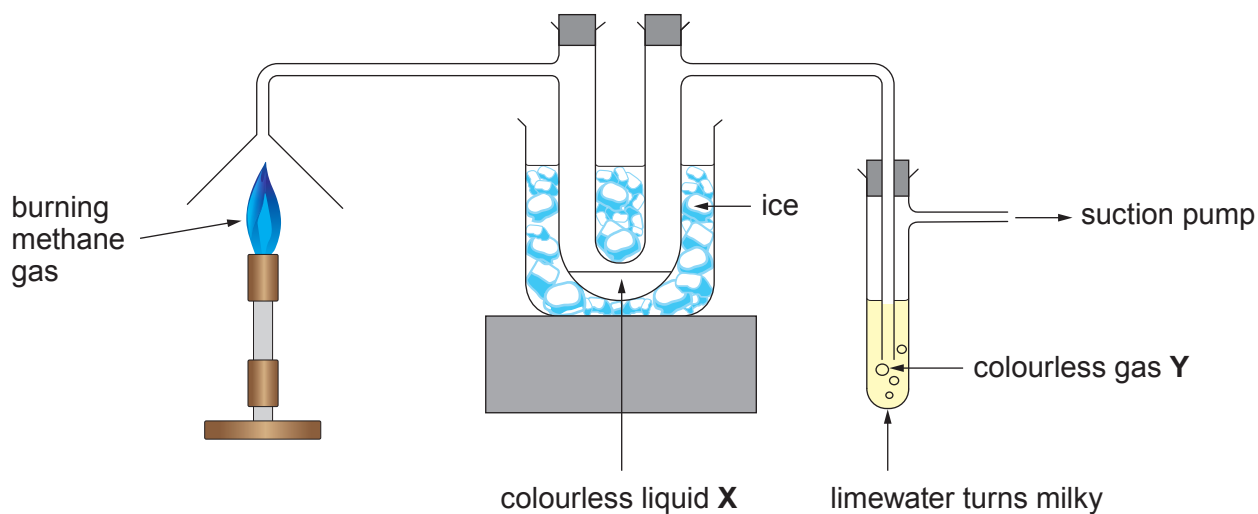
Percentage = %

Examiner
only

6



3. (a) The diagram shows apparatus that can be used to investigate the products formed when methane gas burns. Methane has the formula CH_4 .



- (i) Name the gas present in the air that is needed for methane to burn. [1]

.....

- (ii) Name the products formed. [2]

Colourless liquid **X**

Colourless gas **Y**

- (iii) Tick (✓) the box that gives the name of the group of substances that methane belongs to. [1]

alkenes

☐

monomers

☐

polymers

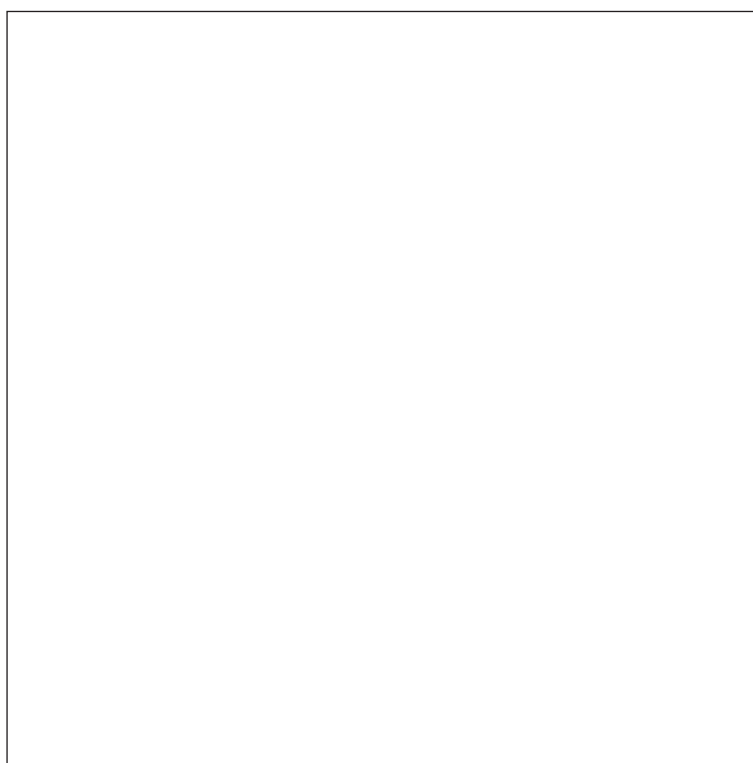
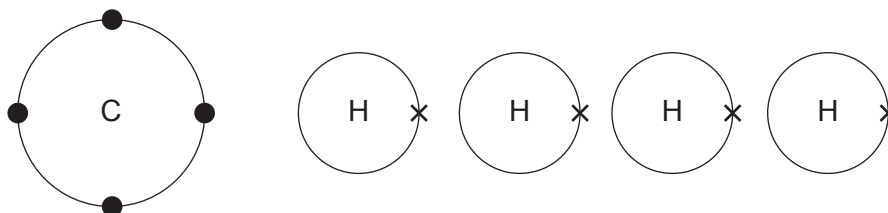
☐

alkanes

☐

- (b) Methane has the formula CH_4 . Its molecules consist of one carbon atom and four hydrogen atoms bonded together.

- (i) Draw a diagram in the box to show how the atoms bond to form a methane molecule. [2]



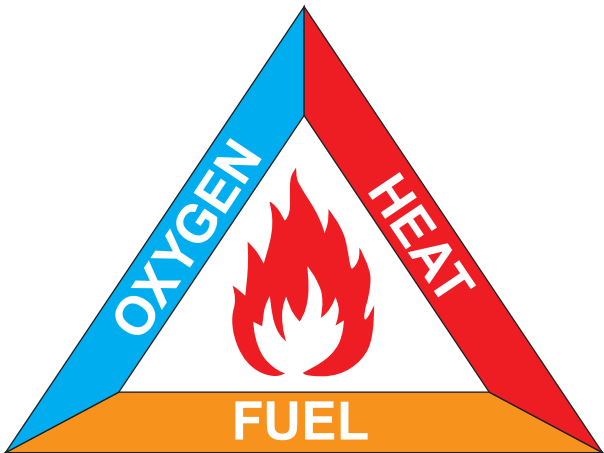
- (ii) Give the name of this type of bonding.

[1]

.....



(c) The fire triangle can be used to explain how fires are extinguished.



The following table gives information about three different types of fire and the methods used to extinguish them.

However, each of the three fires has an **error** in **one** of the columns.

Fire	Type of fire	Firefighting method	How method works
1	chip pan fire	tea towel	removes the heat
2	bonfire	fire blanket	removes the heat
3	electrical fire	fire breaks	removes the fuel

Circle the errors for fires 1 and 2 and then correct them in the table below.

The error for fire 3 has already been circled and the correction given.

[2]

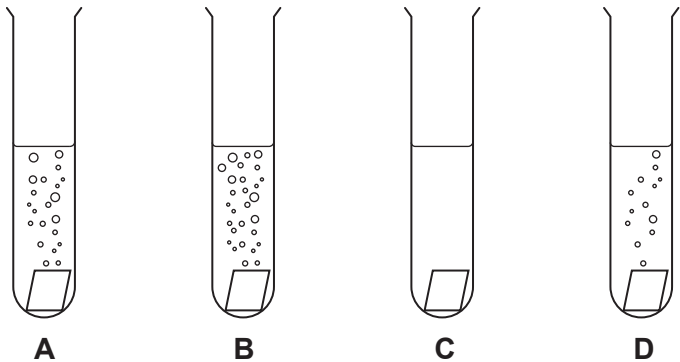
Fire	Correction
1	
2	
3	Fire breaks are used to extinguish a forest fire



3. Nathan and Simon were investigating the reactivity of metals.

- (a) In one experiment, they investigated the effect of adding dilute hydrochloric acid to four different metals, **A**, **B**, **C** and **D**.

The diagram shows the observations made.



- (i) They used the same **volume** of acid in each tube. Give **two other** ways they controlled the dilute hydrochloric acid to ensure a fair test. [2]

1.
2.

- (ii) Use the observations to list the metals in order of their reactivity. Give the reason for your choice. [2]

Most reactive

.....

.....

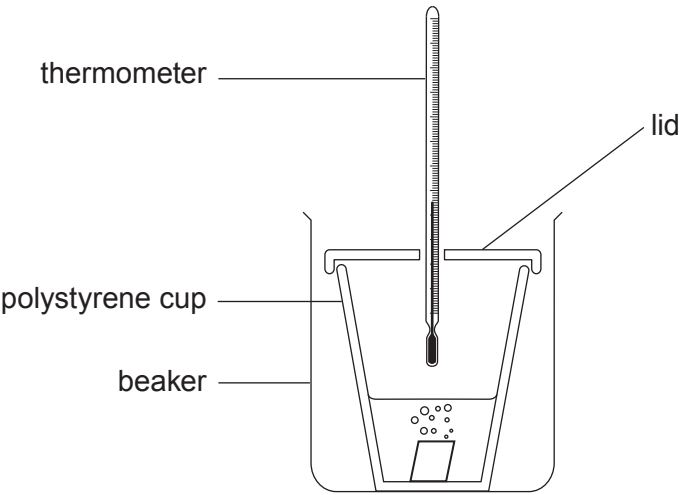
Least reactive

Reason

.....



(b) In a second experiment, Nathan and Simon used the following apparatus to investigate the temperature rise when metals **A**, **B**, **C** and **D** were added to hydrochloric acid.



(i) Identify the error in the way the apparatus is set up and state how this would affect the results collected. [2]

Error

Effect

.....

3430U501
09



(ii) Once they had set up their apparatus correctly, they obtained the following results.

Metal	Temperature rise (°C)			
	Test 1	Test 2	Test 3	Mean
A	22	21		22
B	40	38	36	38
C	0	0	0	0
D	15	9	13	14

I. Identify the anomalous result from metals **B**, **C** and **D**. [1]

Metal Test

II. Calculate the missing result for metal **A**. [1]

Temperature rise for test 3 = °C

III. Zinc is more reactive than metals **C** and **D** but less reactive than metals **A** and **B**. Use this information to predict the temperature rise that would be expected for zinc. [1]

Temperature rise for zinc °C



- (c) The reaction of a metal with an acid is summarised by the following equation.



- (i) Complete the equation for the reaction between magnesium and hydrochloric acid by
- giving the formula of the salt magnesium chloride,
 - balancing the equation.

[2]



- (ii) Give the name of the salt formed when magnesium reacts with nitric acid.

[1]

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